$$
\begin{aligned}
&C_\phi^{(11)\ (22)\ (21)} \to \frac{1}{16\pi^2}\\
&\Big(\frac{3}{2} m_1 \tilde{\mu} g_1^6 LF_{3,3,-2}[m_1,\tilde{\mu}] + \frac{1}{2}\tilde{\mu} g_1^6 m_1^3 LF_{3,3,-1}[m_1,\tilde{\mu}] + \frac{1}{3} m_2 \tilde{\mu} g_2^4 \left(-g_1^2+g_2^2\right) LF_{2,2,0}[m_2,\tilde{\mu}] +\\
&\frac{1}{6} m_2 \tilde{\mu} g_2^4 \left(g_1^2-g_2^2\right) LF_{3,2,-1}[m_2,\tilde{\mu}] + \frac{11}{2} m_2 \tilde{\mu} g_2^6 LF_{3,3,-2}[m_2,\tilde{\mu}] +\\
&\frac{1}{2}\tilde{\mu} g_2^6 m_2^3 LF_{3,3,-1}[m_2,\tilde{\mu}] - \frac{1}{12}\tilde{\mu} g_1^2 y_d^{pr}\overline{a_d}^{pr}\left(g_1^2+3g_2^2\right) LF_{2,2,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] -\\
&\frac{1}{6}\tilde{\mu} g_1^4 y_d^{pr}\overline{a_d}^{pr} LF_{3,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] + \frac{1}{4}\tilde{\mu} g_1^2 y_e^{pr}\overline{a_e}^{pr}\left(g_1^2-g_2^2\right) LF_{2,2,0}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] -\\
&\frac{1}{2}\tilde{\mu} g_1^4 y_e^{pr}\overline{a_e}^{pr} LF_{3,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] + \frac{1}{2}\tilde{\mu} g_1^2 g_2^2 y_e^{pr}\overline{a_e}^{pr} LF_{3,1,0}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] -\\
&\frac{1}{8}\tilde{\mu} y_e^{pr}\overline{a_e}^{pr}\left(3g_1^4+7g_1^2g_2^2-4g_2^4\right) LF_{4,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] +\\
&\frac{1}{12}\tilde{\mu} y_e^{pr}\overline{a_e}^{pr}\left(3g_1^4+8g_1^2g_2^2-5g_2^4\right) LF_{5,1,-2}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] +\\
&\frac{1}{6}\tilde{\mu} g_1^2 y_d^{pr}\overline{a_d}^{pr}\left(2g_1^2+3g_2^2\right) LF_{3,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] - \frac{3}{8}\tilde{\mu} y_d^{pr}\overline{a_d}^{pr}\left(3g_1^4+7g_1^2g_2^2-4g_2^4\right)\\
&\quad LF_{4,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] + \frac{1}{4}\tilde{\mu} y_d^{pr}\overline{a_d}^{pr}\left(3g_1^4+8g_1^2g_2^2-5g_2^4\right) LF_{5,1,-2}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] +\\
&\frac{1}{12}\tilde{\mu} y_u^{pr}\overline{a_u}^{pr}\left(2g_1^4-9g_1^2g_2^2+3g_2^4\right) LF_{2,2,0}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] -\\
&\frac{1}{24}\tilde{\mu} y_u^{pr}\overline{a_u}^{pr}\left(g_1^4-6g_1^2g_2^2-9g_2^4\right) LF_{3,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] + \frac{1}{8}\tilde{\mu} g_2^2 y_u^{pr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right)\\
&\quad LF_{3,2,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \frac{1}{24}\tilde{\mu} y_u^{pr}\overline{a_u}^{pr}\left(7g_1^4-24g_1^2g_2^2+15g_2^4\right) LF_{3,1,0}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] +\\
&\frac{1}{4}\tilde{\mu} g_2^2 y_u^{pr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right) LF_{3,2,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] - \frac{3}{8}\tilde{\mu} y_u^{pr}\overline{a_u}^{pr}\left(3g_1^4+8g_1^2g_2^2-5g_2^4\right)\\
&\quad LF_{4,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] + \frac{1}{4}\tilde{\mu} y_u^{pr}\overline{a_u}^{pr}\left(3g_1^4+8g_1^2g_2^2-5g_2^4\right) LF_{5,1,-2}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] +\\
&\frac{1}{8} m_1\tilde{\mu} g_1^2\left(g_1^4+2g_1^2g_2^2-g_2^4\right) LF_{3,1,0}[\tilde{\mu},m_1] + \frac{3}{4} m_1\tilde{\mu} g_1^4\left(g_1^2-g_2^2\right) LF_{3,2,-1}[\tilde{\mu},m_1] -\\
&\frac{1}{8} m_1\tilde{\mu} g_1^2\left(3g_1^4+7g_1^2g_2^2-4g_2^4\right) LF_{4,1,-1}[\tilde{\mu},m_1] + \frac{1}{2} m_1\tilde{\mu} g_1^4\left(-g_1^2+g_2^2\right) LF_{4,2,-2}[\tilde{\mu},m_1] +\\
&\frac{1}{12} m_1\tilde{\mu} g_1^2\left(3g_1^4+8g_1^2g_2^2-5g_2^4\right) LF_{5,1,-2}[\tilde{\mu},m_1] +\\
&\frac{1}{24} m_2\tilde{\mu} g_2^2\left(9g_1^4+26g_1^2g_2^2-17g_2^4\right) LF_{3,1,0}[\tilde{\mu},m_2] + \frac{31}{12} m_2\tilde{\mu} g_2^4\left(g_1^2-g_2^2\right) LF_{3,2,-1}[\tilde{\mu},m_2] +\\
&\frac{1}{8} m_2\tilde{\mu} g_2^2\left(-9g_1^4-25g_1^2g_2^2+16g_2^4\right) LF_{4,1,-1}[\tilde{\mu},m_2] +\\
&2 m_2\tilde{\mu} g_2^4\left(-g_1^2+g_2^2\right) LF_{4,2,-2}[\tilde{\mu},m_2] + \frac{1}{4} m_2\tilde{\mu} g_2^2\left(3g_1^4+8g_1^2g_2^2-5g_2^4\right) LF_{5,1,-2}[\tilde{\mu},m_2] +\\
&\frac{1}{4}\tilde{\mu} g_1^2 g_2^2\left(g_1^2(7m_1+5m_2)-g_2^2(5m_1+7m_2)\right) LF_{2,2,1,-1}[m_2,\tilde{\mu},m_1] -\\
&\tilde{\mu} g_1^2 g_2^2\left(g_1^2-g_2^2\right)(m_1+m_2) LF_{3,2,1,-2}[m_2,\tilde{\mu},m_1] +\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr} LF_{2,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] +\\
&\frac{3}{8}\tilde{\mu}\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{d}}^t\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr} LF_{2,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr} LF_{2,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] -\\
&\frac{3}{8}\tilde{\mu}\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{d}}^r\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr} LF_{2,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] +\\
&\frac{1}{8}\tilde{\mu}\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{e}}^t\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr} LF_{2,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr} LF_{2,1,1,0}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] -\\
&\frac{1}{8}\tilde{\mu}\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{e}}^r\right] +\\
&\frac{1}{4}\tilde{\mu}\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(-g_1^2+g_2^2\right) LF_{2,1,1,0}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{e}}^t\right] -\\
&\frac{1}{2}\tilde{\mu} g_2^2\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr} LF_{2,1,1,0}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] +\\
&\frac{1}{4}\tilde{\mu}\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(-g_1^2+g_2^2\right) LF_{3,1,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] +\\
&\frac{1}{8}\tilde{\mu}\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{l}}^s, m_{\tilde{e}}^r\right] -\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr} LF_{2,1,1,0}\left[m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] +\\
&\frac{1}{4}\tilde{\mu}\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(g_1^2+g_2^2\right) LF_{3,1,1,-1}\left[m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] +\\
&\frac{1}{4}\tilde{\mu}\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(g_1^2+3g_2^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{d}}^t\right] +\\
&\frac{1}{2}\tilde{\mu}\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(2g_1^2-3g_2^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] +\\
&\frac{3}{4}\tilde{\mu}\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(-g_1^2+g_2^2\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] -\\
&3\tilde{\mu} g_2^2 y_d^{sr}\overline{y_u}^{st} y_u^{pt}\overline{a_d}^{pr} LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{u}}^t\right] + \frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr}\overline{y_u}^{st} y_u^{pt}\overline{a_d}^{pr}\\
&\quad LF_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{u}}^t\right] - \frac{3}{2}\tilde{\mu} g_2^2\overline{y_d}^{st} y_d^{pt} y_u^{sr}\overline{a_u}^{pr} LF_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{u}}^r\right] +\\
&\frac{3}{8}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\left(g_1^2+g_2^2\right)\left(\overline{y_d}^{st} y_d^{pt}-\overline{y_u}^{st} y_u^{pt}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r\right] +\\
&\frac{1}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(3\overline{y_d}^{st} y_d^{pt}\left(-g_1^2+g_2^2\right)+\overline{y_u}^{st} y_u^{pt}\left(g_1^2-9g_2^2\right)\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] +\\
&\frac{3}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right)\left(\overline{y_d}^{st} y_d^{pt}-\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] -\\
&\frac{3}{2}\tilde{\mu} g_2^2\overline{y_d}^{st} y_d^{pt} y_u^{sr}\overline{a_u}^{pr} LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{d}}^t\right] +\\
&\frac{3}{16}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(\overline{y_d}^{st} y_d^{pt}\left(g_1^2-3g_2^2\right)+\overline{y_u}^{st} y_u^{pt}\left(g_1^2+g_2^2\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s\right] -\\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{pt} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-3g_2^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] +\\
&\frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr}\overline{y_u}^{st} y_u^{pt}\overline{a_d}^{pr} LF_{2,2,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{d}}^r\right] -\\
&\frac{1}{2}\tilde{\mu} g_1^2\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr} LF_{2,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] +\\
&\frac{3}{4}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\left(\overline{y_d}^{st} y_d^{pt}\left(g_1^2+g_2^2\right)-2g_2^2\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] -\\
&\frac{1}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2+3g_2^2\right)\left(3\overline{y_d}^{st} y_d^{pt}-\overline{y_u}^{st} y_u^{pt}\right) LF_{2,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] +\\
&\frac{3}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right)\left(\overline{y_d}^{st} y_d^{pt}-\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] +\\
&\frac{3}{16}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(\overline{y_d}^{st} y_d^{pt}\left(g_1^2+g_2^2\right)+\overline{y_u}^{st} y_u^{pt}\left(-3g_1^2+g_2^2\right)\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p\right] +\\
&\frac{1}{4}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(3g_2^2\overline{y_d}^{st} y_d^{pt}+\overline{y_u}^{st} y_u^{pt}\left(4g_1^2-3g_2^2\right)\right) LF_{2,1,1,0}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] +\\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr}\overline{a_u}^{pr}\left(-\overline{y_d}^{st} y_d^{pt}+\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] +\\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{pt} y_u^{sr}\overline{a_u}^{pr}\left(7g_1^2+3g_2^2\right) LF_{2,1,1,0}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] -\\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{pt} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2+g_2^2\right) LF_{3,1,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] +\\
&\frac{1}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{pt} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-3g_2^2\right) LF_{2,1,1,0}\left[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] +\\
&\frac{3}{4}\tilde{\mu}\overline{y_u}^{st} y_u^{pt} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2+g_2^2\right) LF_{3,1,1,-1}\left[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] -\\
&\frac{1}{2} m_1\tilde{\mu} g_1^2 g_2^2\left(g_1^2+g_2^2\right) LF_{2,1,1,0}[\tilde{\mu},m_1,m_2] +\\
&\frac{1}{8}\tilde{\mu} g_1^2 g_2^2\left(g_2^2(11m_1-17m_2)+g_1^2(11m_1+7m_2)\right) LF_{3,1,1,-1}[\tilde{\mu},m_1,m_2] +\\
&\frac{1}{2}\tilde{\mu} g_1^4 g_2^2(2m_1+m_2) LF_{3,2,1,-2}[\tilde{\mu},m_1,m_2] + \frac{1}{2} m_1\tilde{\mu} g_1^4 g_2^2\left(m_1 m_2+8\tilde{\mu}^2\right) LF_{3,2,1,-1}[\tilde{\mu},m_1,m_2] -\\
&\frac{3}{4}\tilde{\mu} g_1^2 g_2^2\left(g_2^2(m_1-3m_2)+g_1^2(m_1+m_2)\right) LF_{4,1,1,-2}[\tilde{\mu},m_1,m_2] +\\
&\frac{1}{2}\tilde{\mu} g_1^2 g_2^2\left(-2g_1^2(m_1+m_2)+g_2^2(5m_1+6m_2)\right) LF_{3,2,1,-2}[\tilde{\mu},m_2,m_1] +\\
&\frac{1}{2} m_2\tilde{\mu} g_1^2 g_2^4\left(-3m_1 m_2+8\tilde{\mu}^2\right) LF_{3,2,1,-1}[\tilde{\mu},m_2,m_1] +\\
&\frac{1}{2}\tilde{\mu} g_1^2 y_d^{sr}\overline{a_d}^{pr}\left(\overline{a_d}^{st} a_d^{pt}-\tilde{\mu}^2\overline{y_d}^{st} y_d^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] +\\
&3\tilde{\mu}\overline{y_d}^{st} y_d^{pt} y_d^{ur}\overline{y_u}^{uv} y_u^{sv}\overline{a_d}^{pr} LF_{1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^v\right] -\\
&\frac{3}{8}\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] +\\
&3\tilde{\mu}\overline{y_d}^{st} y_d^{sr} y_d^{ut}\overline{y_u}^{uv} y_u^{pv}\overline{a_d}^{pr} LF_{1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u\right] +\\
&3\tilde{\mu} y_d^{sr}\overline{y_u}^{st}\overline{y_u}^{uv} y_u^{pv} y_u^{ut}\overline{a_d}^{pr} LF_{1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^u, m_{\tilde{u}}^t\right] -\\
&\frac{3}{8}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\overline{a_d}^{st} a_d^{pt}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2 y_d^{sr}\overline{a_d}^{pr}\left(\overline{a_d}^{st} a_d^{pt}-\tilde{\mu}^2\overline{y_d}^{st} y_d^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] +\\
&\frac{3}{8}\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] +\\
&3\tilde{\mu}\overline{y_d}^{st} y_d^{ut}\overline{y_u}^{uv} y_u^{pv} y_u^{sr}\overline{a_u}^{pr} LF_{1,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^v\right] +\\
&\frac{3}{8}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\overline{a_d}^{st} a_d^{pt}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] +\\
&3\tilde{\mu}\overline{y_d}^{st}\overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_u^{ur}\overline{a_u}^{pr} LF_{1,1,1,1,0}\left[m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2 y_e^{sr}\overline{a_e}^{pr}\left(\overline{a_e}^{st} a_e^{pt}-\tilde{\mu}^2\overline{y_e}^{st} y_e^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] -\\
&\frac{1}{8}\tilde{\mu}^3\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] -\\
&\frac{1}{8}\tilde{\mu} y_e^{sr}\overline{a_e}^{pr}\overline{a_e}^{st} a_e^{pt}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^r, m_{\tilde{l}}^s, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] +\\
&\frac{1}{2}\tilde{\mu} g_1^2 y_e^{sr}\overline{a_e}^{pr}\left(\overline{a_e}^{st} a_e^{pt}-\tilde{\mu}^2\overline{y_e}^{st} y_e^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s\right] +\\
&\frac{1}{8}\tilde{\mu}^3\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{l}}^s\right] +\\
&\frac{1}{8}\tilde{\mu} y_e^{sr}\overline{a_e}^{pr}\overline{a_e}^{st} a_e^{pt}\left(g_1^2+g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^t, m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{l}}^p\right] +\\
&\frac{1}{4}\tilde{\mu} y_e^{sr}\overline{a_e}^{pr}\left(-2g_2^2\overline{a_e}^{st} a_e^{pt}+\tilde{\mu}^2\overline{y_e}^{st} y_e^{pt}\left(g_1^2-g_2^2\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] +\\
&\frac{1}{4}\tilde{\mu} y_e^{sr}\overline{a_e}^{pr}\overline{a_e}^{st} a_e^{pt}\left(-g_1^2+g_2^2\right) LF_{3,1,1,1,-1}\left[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^s\right] +\\
&\frac{1}{4}\tilde{\mu} y_e^{sr}\overline{a_e}^{pr}\left(\overline{a_e}^{st} a_e^{pt}\left(-g_1^2+g_2^2\right)+2g_1^2\tilde{\mu}^2\overline{y_e}^{st} y_e^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] +\\
&\frac{1}{4}\tilde{\mu}^3\overline{y_e}^{st} y_e^{pt} y_e^{sr}\overline{a_e}^{pr}\left(-g_1^2+g_2^2\right) LF_{3,1,1,1,-1}\left[m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p\right] - \frac{1}{4}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\\
&\quad\left(2\overline{a_d}^{st} a_d^{pt}\left(-2g_1^2+3g_2^2\right)+\tilde{\mu}^2\overline{y_d}^{st} y_d^{pt}\left(g_1^2+3g_2^2\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] +\\
&\frac{3}{4}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\overline{a_d}^{st} a_d^{pt}\left(-g_1^2+g_2^2\right) LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] +\\
&3\tilde{\mu} g_2^2 y_u^{pt}\overline{a_d}^{pr}\left(-a_d^{sr}\overline{a_u}^{st}+\tilde{\mu}^2 y_d^{sr}\overline{y_u}^{st}\right) LF_{2,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] +\\
&\frac{3}{2}\tilde{\mu} g_2^2 y_u^{pt}\overline{a_d}^{pr}\left(a_d^{sr}\overline{a_u}^{st}-\tilde{\mu}^2 y_d^{sr}\overline{y_u}^{st}\right) LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] +\\
&\frac{3}{2} g_2^2\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_u^{sr}\overline{a_u}^{pr} LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] -\\
&\frac{3}{8}\tilde{\mu}\overline{a_d}^{pr}\left(\overline{a_u}^{st}\left(y_d^{sr} a_u^{pt}\left(g_1^2+g_2^2\right)+y_u^{pt} a_d^{sr}\left(g_1^2-3g_2^2\right)\right)+\tilde{\mu}^2 y_d^{sr}\overline{y_u}^{st} y_u^{pt}\left(g_1^2+g_2^2\right)\right)\\
&\quad LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{u}}^t\right] -\\
&\frac{3}{8}\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-3g_2^2\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{u}}^r\right] +\\
&3\tilde{\mu}\overline{y_d}^{st} y_d^{pt}\overline{y_u}^{uv} y_u^{sv} y_u^{ur}\overline{a_u}^{pr} LF_{1,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r\right] + \frac{1}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\\
&\quad\left(\overline{a_u}^{st} a_u^{pt}\left(g_1^2-9g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(5g_1^2-9g_2^2\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] -\\
&\frac{3}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right)\left(\overline{a_u}^{st} a_u^{pt}+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] +\\
&\frac{3}{2} g_2^2\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_u^{sr}\overline{a_u}^{pr} LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s\right] + \frac{3}{16}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\\
&\quad\left(\overline{a_u}^{st} a_u^{pt}\left(g_1^2+g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(-3g_1^2+g_2^2\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] +\\
&\frac{3}{2}\tilde{\mu} g_2^2 y_u^{pt}\overline{a_d}^{pr}\left(a_d^{sr}\overline{a_u}^{st}-\tilde{\mu}^2 y_d^{sr}\overline{y_u}^{st}\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^s\right] -\\
&\frac{3}{16}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(\overline{a_u}^{st} a_u^{pt}\left(g_1^2+g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(g_1^2-3g_2^2\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] +\\
&\frac{1}{4}\tilde{\mu} y_d^{sr}\overline{a_d}^{pr}\left(\overline{a_d}^{st} a_d^{pt}\left(g_1^2+3g_2^2\right)+2g_1^2\tilde{\mu}^2\overline{y_d}^{st} y_d^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] +\\
&\frac{3}{4}\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_d^{sr}\overline{a_d}^{pr}\left(-g_1^2+g_2^2\right) LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p\right] -\\
&\frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] + \frac{1}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\\
&\quad\left(\overline{a_u}^{st} a_u^{pt}\left(g_1^2+3g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(5g_1^2+3g_2^2\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] -\\
&\frac{3}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right)\left(\overline{a_u}^{st} a_u^{pt}+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] + \frac{3}{16}\tilde{\mu}\\
&\quad y_u^{sr}\overline{a_u}^{pr}\left(\overline{a_u}^{st} a_u^{pt}\left(-3g_1^2+g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(g_1^2+g_2^2\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t\right] -\\
&\frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p\right] - \frac{3}{16}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\\
&\quad\left(\overline{a_u}^{st} a_u^{pt}\left(g_1^2-3g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(g_1^2+g_2^2\right)\right) LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r\right] - \frac{1}{4}\tilde{\mu} y_u^{sr}\\
&\quad\overline{a_u}^{pr}\left(\overline{a_u}^{st} a_u^{pt}\left(-4g_1^2+3g_2^2\right)+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\left(4g_1^2+3g_2^2\right)\right) LF_{2,1,1,1,0}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] +\\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr}\overline{a_u}^{pr}\left(\overline{a_u}^{st} a_u^{pt}+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] -\\
&\frac{3}{8}\tilde{\mu} y_u^{sr}\overline{a_u}^{pr}\left(g_1^2-g_2^2\right)\left(\overline{a_u}^{st} a_u^{pt}+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\right) LF_{2,2,1,1,-1}\left[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s\right] -\\
&\frac{1}{4}\tilde{\mu} g_1^2 y_u^{sr}\overline{a_u}^{pr}\left(-7\overline{a_u}^{st} a_u^{pt}+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\right) LF_{2,1,1,1,0}\left[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] -\\
&\frac{3}{4}\tilde{\mu} g_1^2 y_u^{sr}\overline{a_u}^{pr}\left(\overline{a_u}^{st} a_u^{pt}+\tilde{\mu}^2\overline{y_u}^{st} y_u^{pt}\right) LF_{3,1,1,1,-1}\left[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r\right] +\\
&3\tilde{\mu}^3\overline{y_d}^{st}\overline{y_d}^{uv}\overline{a_d}^{pr}\left(y_d^{pt} y_d^{sv} y_d^{ur}+y_d^{pv} y_d^{sr} y_d^{ut}\right) LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^u\right] +\\
&3\tilde{\mu}\overline{a_d}^{pr}\left(y_d^{pt}\overline{y_u}^{uv} y_u^{sv}\overline{a_d}^{st} a_d^{ur}+\tilde{\mu}^2\overline{y_d}^{st}\overline{y_d}^{uv} y_d^{pv} y_d^{sr} y_d^{ut}\right)\\
&\quad LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u\right] +\\
&3\tilde{\mu}\overline{y_u}^{uv}\overline{a_d}^{pr}\left(y_d^{sr} y_d^{ut}\overline{a_d}^{st} a_u^{pv}-\tilde{\mu}^2\overline{y_d}^{st} y_d^{pt} y_d^{ur} y_u^{sv}\right) LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^v\right] -\\
&3\tilde{\mu}^3\overline{y_d}^{st} y_d^{pt} y_d^{ur}\overline{y_u}^{uv} y_u^{sv}\overline{a_d}^{pr} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^u, m_{\tilde{u}}^v\right] +\\
&3\tilde{\mu}\overline{y_d}^{uv} y_d^{pv} y_u^{ut}\overline{a_d}^{pr} a_d^{sr}\overline{a_u}^{st} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t\right] +\\
&3\tilde{\mu}\overline{a_d}^{pr}\left(\overline{a_u}^{st}\left(\overline{y_d}^{uv} y_d^{sv} y_d^{ur} a_u^{pt}+\overline{y_u}^{uv} y_u^{pv} y_u^{ut} a_d^{sr}\right)-\tilde{\mu}^2 y_d^{sr}\overline{y_u}^{st}\overline{y_u}^{uv} y_u^{pv} y_u^{ut}\right)\\
&\quad LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^t\right] +\\
&3\tilde{\mu} y_d^{ur}\overline{y_u}^{uv} y_u^{sv}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{u}}^v\right] -\\
&3\tilde{\mu}^3\overline{y_d}^{st} y_d^{ut}\overline{y_u}^{uv} y_u^{pv} y_u^{sr}\overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^v\right] -\\
&3\tilde{\mu}^3\overline{y_d}^{st} y_d^{ut}\overline{y_u}^{uv} y_u^{pv} y_u^{sr}\overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^v\right] -\\
&3\tilde{\mu}^3\overline{y_d}^{st}\overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_u^{ur}\overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r\right] +\\
&3\tilde{\mu}\overline{y_d}^{uv} y_u^{sr} y_u^{ut} a_d^{pv}\overline{a_u}^{pr}\overline{a_u}^{st} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] +\\
&\tilde{\mu}^3\overline{y_e}^{st}\overline{y_e}^{uv}\overline{a_e}^{pr}\left(y_e^{pt} y_e^{sv} y_e^{ur}+y_e^{pv} y_e^{sr} y_e^{ut}\right) LF_{1,1,1,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{e}}^v, m_{\tilde{l}}^p, m_{\tilde{l}}^u\right] +\\
&\tilde{\mu}^3\overline{y_e}^{st}\overline{y_e}^{uv} y_e^{pv} y_e^{sr} y_e^{ut}\overline{a_e}^{pr} LF_{1,1,1,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{l}}^s, m_{\tilde{l}}^u\right] +\\
&3\tilde{\mu}\overline{y_d}^{uv} y_d^{sv} y_u^{pt}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{ur} LF_{1,1,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] +\\
&3\tilde{\mu}^3\overline{y_u}^{st}\overline{y_u}^{uv} y_u^{pt} y_u^{sv} y_u^{ur}\overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^v\right] +\\
&3\tilde{\mu}^3\overline{y_u}^{st}\overline{y_u}^{uv}\overline{a_u}^{pr}\left(y_u^{pt} y_u^{sv} y_u^{ur}+y_u^{pv} y_u^{sr} y_u^{ut}\right) LF_{1,1,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{u}}^v\right] +\\
&3\tilde{\mu}^3\overline{y_d}^{uv}\overline{a_d}^{pr}\overline{a_d}^{st}\left(y_d^{sv} y_d^{ur} a_d^{pt}+y_d^{sr} y_d^{ut} a_d^{pv}\right) LF_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u\right] +\\
&3\tilde{\mu}\overline{a_d}^{pr}\left(\tilde{\mu}^4\overline{y_d}^{st} y_d^{pt} y_d^{ur}\overline{y_u}^{uv} y_u^{sv}+\overline{a_d}^{st}\left(y_d^{pt} a_d^{ur}\overline{a_u}^{uv} a_u^{sv}-\tilde{\mu}^2 y_d^{sr} y_d^{ut}\overline{y_u}^{uv} a_u^{pv}\right)\right)\\
&\quad LF_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^v\right] -\\
&3\tilde{\mu}^3\overline{y_d}^{uv} y_d^{pv} y_u^{ut}\overline{a_d}^{pr} a_d^{sr}\overline{a_u}^{st} LF_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^t\right] + 3\tilde{\mu}\overline{a_d}^{pr}\\
&\quad\overline{a_u}^{st}\left(y_u^{ut} a_d^{sr}\overline{a_u}^{uv} a_u^{pv}-\tilde{\mu}^2 y_d^{ur}\overline{y_u}^{uv} y_u^{sv} a_u^{pt}\right) LF_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^t, m_{\tilde{u}}^v\right] +\\
&3\tilde{\mu}^5\overline{y_d}^{st} y_d^{ut}\overline{y_u}^{uv} y_u^{pv} y_u^{sr}\overline{a_u}^{pr} LF_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^v\right] -\\
&3\tilde{\mu}^3\overline{y_d}^{uv} y_u^{sr} y_u^{ut} a_d^{pv}\overline{a_u}^{pr}\overline{a_u}^{st} LF_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^v, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^t\right] +\\
&\tilde{\mu}^3\overline{y_e}^{uv}\overline{a_e}^{pr}\overline{a_e}^{st}\left(y_e^{sv} y_e^{ur} a_e^{pt}+y_e^{sr} y_e^{ut} a_e^{pv}\right) LF_{1,1,1,1,1,1,0}\left[m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{e}}^v, m_{\tilde{l}}^p, m_{\tilde{l}}^s, m_{\tilde{l}}^u\right] +\\
&3\tilde{\mu}^3\overline{y_u}^{uv}\overline{a_u}^{pr}\overline{a_u}^{st}\left(y_u^{sv} y_u^{ur} a_u^{pt}+y_u^{sr} y_u^{ut} a_u^{pv}\right) LF_{1,1,1,1,1,1,0}\left[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{q}}^u, m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{u}}^v\right]\Big)
\end{aligned}
$$